

ACC/AHA & JNC-8 Guidelines: Hypertension

1. Classification of Blood Pressure in Adults

Blood pressure categories according to ACC/AHA criteria:

- Normal: Systolic < 120 mmHg AND Diastolic < 80 mmHg.
- Elevated: Systolic 120-129 mmHg AND Diastolic < 80 mmHg.
- Stage 1 Hypertension: Systolic 130-139 mmHg OR Diastolic 80-89 mmHg.
- Stage 2 Hypertension: Systolic $\geq$ 140 mmHg OR Diastolic $\geq$ 90 mmHg.
- Hypertensive Crisis: Systolic > 180 mmHg and/or Diastolic > 120 mmHg.

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2. First-Line Antihypertensive Medications

Initial monotherapy or combination therapy includes four primary classes:

- Thiazide or Thiazide-like Diuretics (Chlorthalidone, Hydrochlorothiazide).
- ACE Inhibitors (Lisinopril, Enalapril) - reduce proteinuria in diabetic nephropathy.
- Angiotensin Receptor Blockers (Losartan, Valsartan) - alternative if ACEi cough occurs.
- Calcium Channel Blockers (Amlodipine, Nifedipine) - potent vasodilators.

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3. Special Population Guidelines

Demographic and comorbid adjustments for treatment selection:

- Black Patients (without CKD): Initial treatment should include a CCB or Thiazide diuretic.
- Chronic Kidney Disease (CKD) with Proteinuria: ACE Inhibitor or ARB is first-line regardless of race.
- Pregnancy: Labetalol, Nifedipine ER, or Methyldopa. ACEi and ARBs are strictly contraindicated.